

ROBOTIC WELDING · ILUO SKILL PATH

Technical Setter Interns

Diego Adair de León Márquez

Universidad Politécnica de Aguascalientes
Mechatronics Engineering · 9th term

Plant tutor: Fernando Robledo

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TENNECO

Our value: WIN

"We must earn the trust of our employees and customers."

» Trust comes before access

On the floor nobody lets you touch a welding cell because of a job title. You get there by being useful first, and by being useful again the next day.

» I learned by answering, not by asking

Most of what I know came from being the one who showed up when a cell went down — not from waiting for a training session to be scheduled.

» Winning means the line runs

My progress only counts if the cell keeps welding. That is the scoreboard I use on myself, and it is the one this plant is paid on.

Trust is the only currency an intern actually has.

Where I work

Robotic GMAW welding — exhaust systems

PRODUCT

Exhaust systems

Clean Air business unit

PROCESS

GMAW

MIG / MAG · robotic cells

ROBOTS

Yaskawa

Most of the cells · one
Fanuc

EQUIPMENT

SKS · Miller

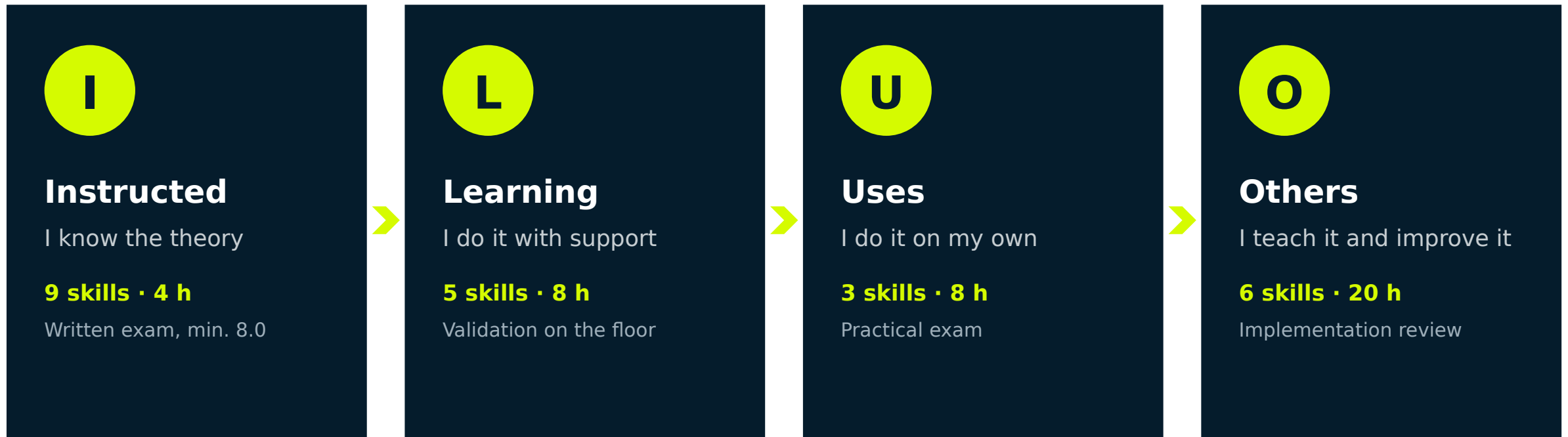
Welding systems and
controls

My zone has few welding cells and they run stable.

That single fact shaped how my learning actually happened — I could not learn by waiting for them to fail.

The path I am expected to walk

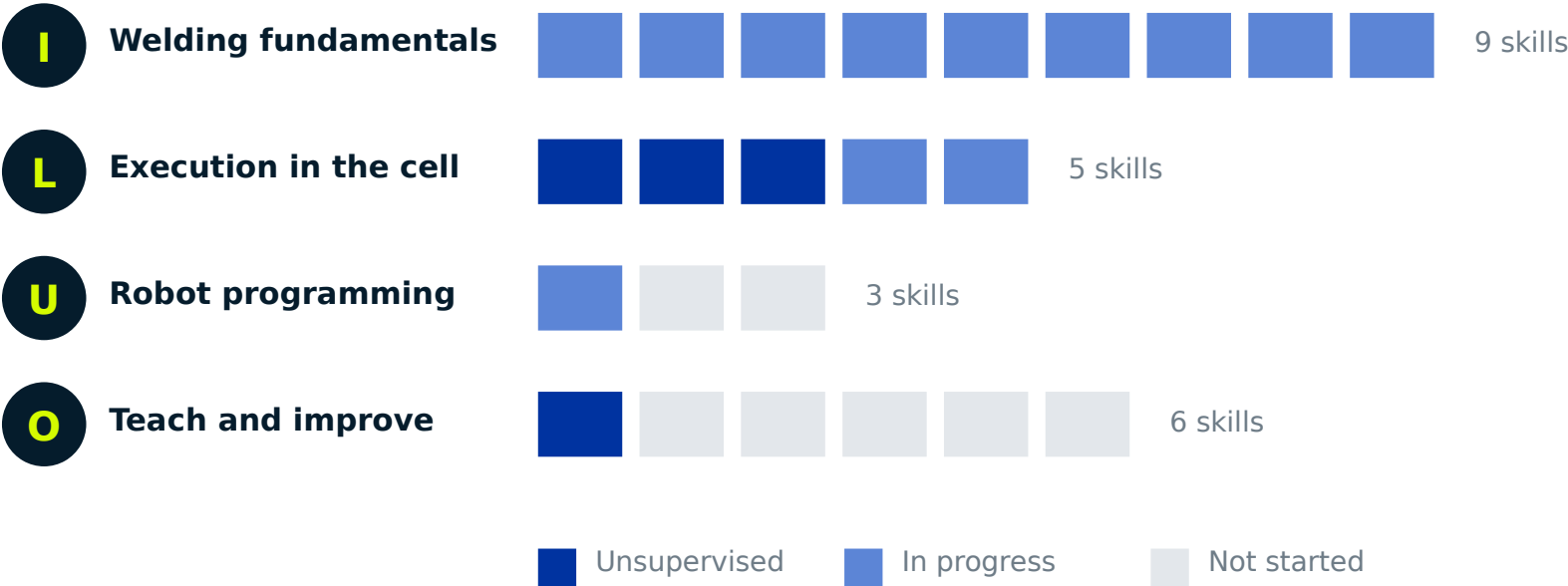
ILUO — 4 levels · 23 skills · 40 hours



Each level assumes the one before it. Half of the total hours sit in level O — this program is built to multiply knowledge, not just to certify individuals.

Where I stand today

4 of 23 skills unsupervised · 12 in progress · 7 not started



WHAT I DO UNSUPERVISED

- Consumable changes — contact tip, liner, nozzle, diffuser, rollers
- Essential variable adjustment — current, wire feed, voltage, travel speed
- Welding symbol interpretation
- Closing ANDON orders

The honest part

Level I content I picked up in the cell, not in a classroom. The theory session and the 8.0 written exam have not been delivered yet — so none of it is formally validated.

The plan and the reality

WHAT THE PLAN ASSUMED

40 scheduled hours

Classroom first, then the cell

One level at a time, in order

A formal exam at every gate

WHAT ACTUALLY HAPPENED

Hours driven by demand, not by schedule

Straight to the floor from day one

Levels touched out of order

No formal evaluation yet

60% integration project | 40% breakdowns and support

HOW MY TIME REALLY SPLIT

Most of my hours went into an integration project with my engineer — work that is not on the welding matrix at all.

This is not a gap in effort. It is a gap in structure — and it is fixable before January.

The integration project

Where 60% of my time went — and none of it is on the ILUO matrix

WHAT IT IS

[describe the integration in one line]

MY ROLE

[what you own day to day]

WHAT IT TAUGHT ME

[three skills outside the welding matrix]

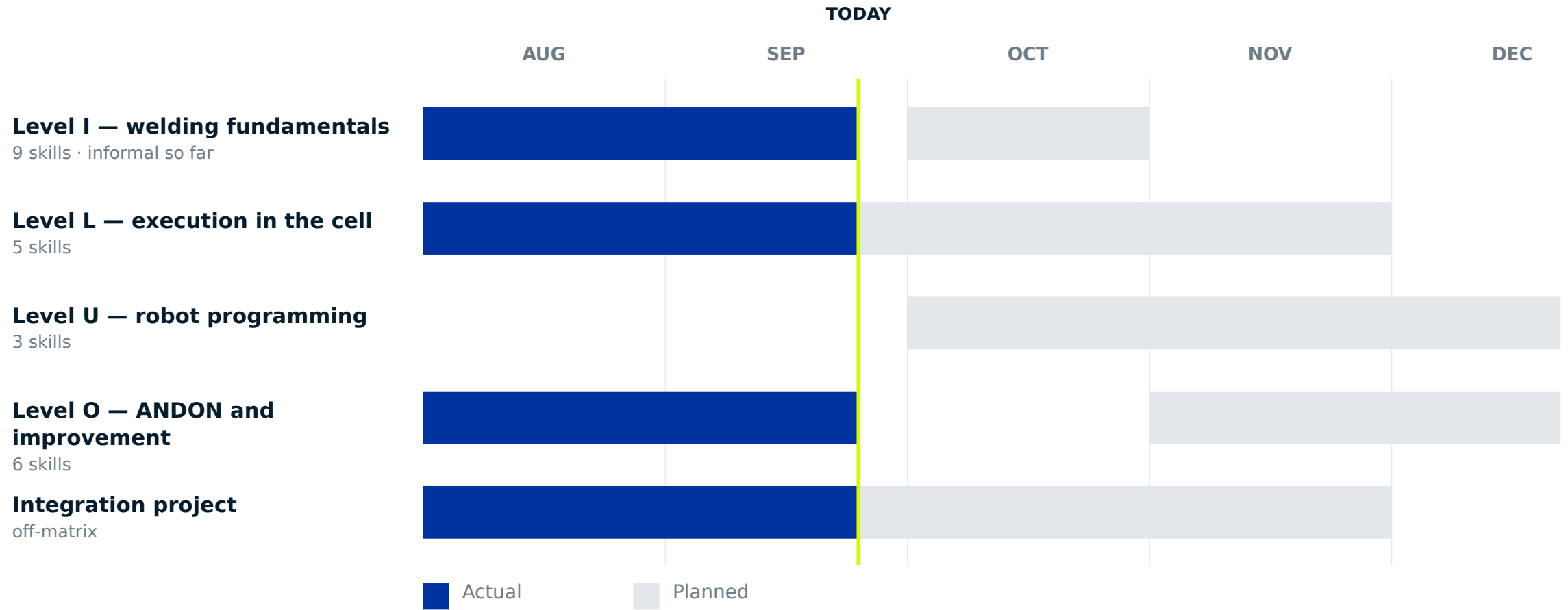
STATUS

[% complete · next milestone]

Worked directly with my engineering lead — this is the experience the matrix does not capture.

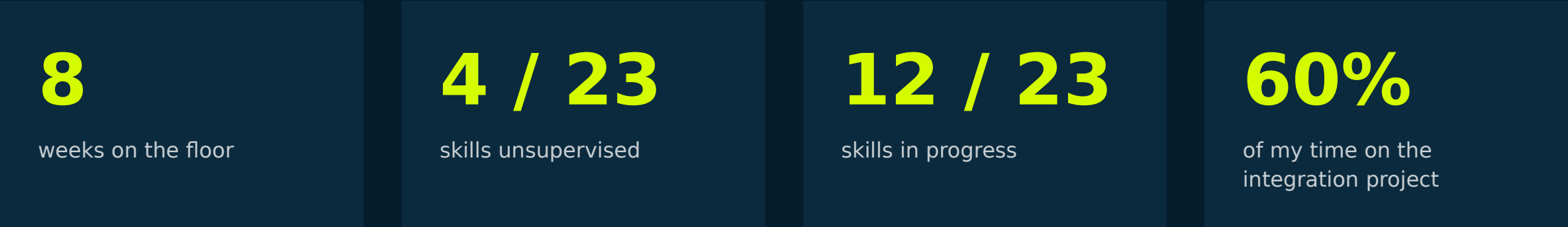
Timeline

August 1, 2026 → January 1, 2027



14 weeks left. Levels U and O are the ones still fully ahead of me.

Results to date



WHAT CHANGED IN EIGHT WEEKS

Consumable change	I watched someone do it	>>	I do it alone
Welding symbols	I could not read a joint callout	>>	I read the print and find the joint
ANDON	I was the one who called it in	>>	I am the one who closes it

What is missing, and how I close it

Four commitments before January 1

COMMITMENT	WHAT IT UNLOCKS	WHEN	WHAT I NEED
1 Sit the Level I theory session and the written exam	Level I formally validated	October	Two half-days scheduled
2 Document 10 parameter validations against the WPS	Closes Level L	Oct - Nov	Access to parameter sheets
3 Create one welding program in SKS, supervised	Opens Level U	November	Cell time and supervision
4 Take one improvement idea to implementation	First Level O evidence	Nov - Dec	A sponsor for the idea

None of this needs budget. It needs calendar.

Conclusions

01 I am real at Level L, and honest about Level I.

Four skills I execute with nobody watching. But nothing is formally validated yet, and that is the gap that actually matters.

02 The floor taught me faster than the schedule would have.

Demand-driven learning built troubleshooting judgment a classroom does not give you. It just did not produce paperwork.

03 Fourteen weeks left, and a plan for them.

The four commitments take me from informally capable to formally certified before January 1.

Trust first, then the torch. That is how I read WIN.

Thank you

Questions?

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